Xplorer V1.1 IDEX — BTT Octopus V1.1 wiring guide

Printable bench copy · generated from Octopus_V1.1_Wiring.md · 2026-09-02

Replaces the BTT Manta M8P V2.0 + CB2 with a **BTT Octopus V1.1 (STM32F446)** driven over USB by a **Raspberry Pi 5**. All seven mainboard motors run **TMC2240 in SPI mode**. The two EBB toolboards keep their onboard TMC2209 extruder drivers and are not touched.

Everything below is derived from the config in `_deploy/` — if you change the wiring, the config has to change with it.

1. What moves and what does not

SUBSYSTEM	BEFORE	AFTER
Mainboard	Manta M8P V2.0	Octopus V1.1
Host	CB2 (on the Manta)	Raspberry Pi 5
Host ↔ mainboard link	UART `/dev/ttyAMA0`	USB-C
X, X1, Y, Y1, Z, Z1, Z2 drivers	TMC2209 UART	**TMC2240 SPI**
T0/T1 extruder drivers	TMC2209 on the EBB	unchanged
Hotends, part fans, X endstops, smart sensors	on the EBB toolboards	unchanged
Cartographer V4	USB	USB, now to the Pi
Nozzle alignment camera	USB	USB, now to the Pi

Only 25 mainboard signals exist in this build. Everything on the toolhead is on the toolboards and needs no rework.

2. Parts and consumables

- 7 × TMC2240 stepper driver modules (SPI-capable, e.g. BTT TMC2240 V1.0 — `rref` 12 kΩ)
- Driver heatsinks, plus airflow over the driver bank
- A pile of 2.54 mm jumper caps: SPI jumpers for 7 driver slots, plus 6 fan voltage selectors
- USB-C cable, Pi → Octopus
- Powered USB 2.0 hub (you already have a VIA Labs hub in the machine)
- Official Raspberry Pi 5 27 W USB-C PSU

3. Power terminals

The Octopus has **four** screw terminals. Feeding all inputs from one 24 V PSU with a common ground is fine.

TERMINAL	CONNECT	NOTES
`PWR IN`	24 V from PSU	Logic, fans, hotend FETs. Must be **> 14.1 V** — do not feed 12 V here
`MOTOR PWR`	24 V from PSU	Supplies all eight driver slots. TMC2240 is happy at 24 V
`BED IN`	24 V from PSU	Bed heater supply, up to 300 W

TERMINAL	CONNECT	NOTES
`BED OUT`	to the bed heater pads	Switched by `PA1`

Polarity is silkscreened on the **underside** of the board. Getting it wrong kills the board.

> **MCU power jumper:** there is a jumper that lets the board run off USB-C alone, for DFU flashing. Leave it **off** in normal operation and never fit it while the PSU is connected.

4. Driver slots and jumpers

4.1 Jumpers — do this before inserting any driver

1. **Remove every jumper under all eight driver slots.** Boards ship with UART or microstep jumpers fitted, and leftovers will short the SPI bus.
2. Fit the **SPI mode** jumper pattern under slots **0 through 6**. This is section 3.3 of the BTT Octopus user guide ("3.3 SPI MODE") — the pattern is only given as an image, so follow the manual picture rather than any text description. It bridges each driver's `SW_MOSI` / `SW_MISO` / `SW_SCK` pins onto the shared SPI1 bus (`PA7` / `PA6` / `PA5`).
3. Leave **slot 7** empty with no jumpers.
4. **Remove all DIAG jumpers** (section 4.2 of the manual). Those tie a driver's stallguard output onto the very same `PG9` – `PG15` inputs we use for physical endstops. A single forgotten DIAG jumper will make an endstop read permanently triggered.

You cannot mix these SPI drivers with a step/dir driver at anything other than 16 microsteps — the microstep pins are the SPI pins. Not an issue here since every slot is SPI or empty.

4.2 Slot → axis → motor connector

Drivers all face the same way; line the module's `EN` pin up with the `EN` marking on the socket silkscreen.

SLOT	KLIPPER STEPPER	PHYSICAL AXIS	MOTOR CONNECTOR	CS PIN
DRIVER0	`stepper_x`	X, T0 carriage	`MOTOR0`	`PC4`
DRIVER1	`dual_carriage`	X1, T1 carriage	`MOTOR1`	`PD11`
DRIVER2	`stepper_y`	Y, right side	**`MOTOR2_1`**	`PC6`
DRIVER3	`stepper_y1`	Y1, left side	`MOTOR3`	`PC7`
DRIVER4	`stepper_z`	Z, front-left	`MOTOR4`	`PF2`
DRIVER5	`stepper_z1`	Z1, back	`MOTOR5`	`PE4`
DRIVER6	`stepper_z2`	Z2, front-right	`MOTOR6`	`PE1`
DRIVER7	—	spare	`MOTOR7`	`PD3`

> **The `MOTOR2_1` / `MOTOR2_2` trap.** The Octopus has nine motor connectors but only eight drivers: `MOTOR2_1` and `MOTOR2_2` are wired **in parallel off DRIVER2**. Plug Y into `MOTOR2_1` and leave `MOTOR2_2` **empty**. Anything plugged into `MOTOR2_2` moves in lockstep with Y.

4.3 Motor cable pinout

Each header is a 4-pin JST-XH with the coil pairs silkscreened next to it. If you are reusing the Manta harnesses, verify before plugging in: with the motor unplugged, buzz the four wires for continuity — the two that read a few ohms to each other are one coil, and each coil must land on one silkscreened pair.

If a motor stutters and grinds, the coils are split across the pairs. If a motor simply runs the wrong way, **do not rewire it** — flip the `!` on that stepper's `dir_pin` in the config.

5. Endstops and switch inputs

All eight of these are simple switch-to-GND connections: **signal + GND**. Where a header has a third pin it is a supply rail for powered sensors, which none of these need.

HEADER	PIN	CONNECT	CONFIG
`X-STOP`	`PG6`	*nothing*	X and X1 home against switches on the EBB toolboards
`Y-STOP`	`PG9`	Y endstop switch	`endstop_pin: ~PG9`
`Z-STOP`	`PG10`	Z endstop, front-left	`endstop_pin: PG10`
`Z2-STOP`	`PG11`	Z1 endstop, back	`endstop_pin: PG11`
`E0DET`	`PG12`	Z2 endstop, front-right	`endstop_pin: PG12`
`E1DET`	`PG13`	tool offset contact probe	`pin: ^PG13`
`E2DET`	`PG14`	T0 (left) buffer runout switch	`switch_pin: ^PG14`
`E3DET`	`PG15`	T1 (right) buffer runout switch	`switch_pin: ^PG15`

The three Z switches are physically identical, so label the cables — swapping two of them will make `Z_TILT_ADJUST` drive the gantry into a corner.

The dedicated `PROBE` port (`PB7`) stays empty; probing is done by the Cartographer over USB.

6. Thermistors

Non-polarised, either way round.

HEADER	PIN	CONNECT	SENSOR TYPE IN CONFIG
`TB`	`PF3`	bed thermistor	`Generic 3950`
`T0` / `TH0`	`PF4`	*nothing*	hotend thermistors are on the toolboards
`T1` / `TH1`	`PF5`	chamber thermistor	`EPCOS 100K B57560G104F`
`T2` / `TH2`	`PF6`	*nothing*	
`T3` / `TH3`	`PF7`	*nothing*	

7. Heater outputs

TERMINAL	PIN	CONNECT
`BED OUT` / `HB`	`PA1`	bed heater pads
`HE0`	`PA2`	*nothing*

TERMINAL	PIN	CONNECT
`HE1`	`PA3`	*nothing*
`HE2`	`PB10`	*nothing*
`HE3`	`PB11`	*nothing*

Both hotends are driven by their own EBB toolboards, so all four hotend outputs stay unused.

8. Fan headers and switched outputs

Set the voltage selector jumper on each header before you plug anything in. Every fan header has its own 5 V / 12 V / 24 V selector, and the wrong setting will destroy whatever is attached. Copy the setting each device had on the Manta.

HEADER	PIN	CONNECT	VOLTAGE JUMPER
`FAN0`	`PA8`	case light LED strip (PWM)	match the strip
`FAN1`	`PE5`	*free*	—
`FAN2`	`PD12`	chamber fan	match the fan
`FAN3`	`PD13`	electronics bay fan	match the fan
`FAN4`	`PD14`	T0 (left) LLL buffer supply	match the buffer
`FAN5`	`PD15`	T1 (right) LLL buffer supply	match the buffer
always-on ×2	—	*free*	—

Polarity matters, and BTT swapped the fan polarity silkscreen on some early boards — check against PINS.pdf rather than trusting the underside marking.

Aim the electronics bay fan across the driver bank. Y and Y1 are set to 1.5 A RMS and Z to 1.4 A; TMC2240 at those currents needs both its heatsink and moving air.

PS-ON

HEADER	PIN	CONNECT
`PS-ON`	`PE11`	PSU relay / SSR control line — `[output_pin Power]`

This is the board's dedicated PS-ON port, the same function `PD14` served on the Manta M8P V2.0, so `[output_pin Power]` carries over unchanged — `value: 1` still means "PSU on" and the relay wiring is identical. Move the relay control lead across as-is.

The one board-specific piece is in the firmware build, not the config: the Octopus `.config` sets `CONFIG_INITIAL_PINS="PE11"`, mirroring `"PD14"` on the Manta. That drives PS-ON high the moment the MCU boots, so the PSU stays up through a `FIRMWARE_RESTART` instead of cutting out before Klipper loads its config.

9. USB and the Raspberry Pi 5

Six USB devices, and the Pi 5 has four ports. Known identities from the live system:

DEVICE	USB ID	WHERE TO PLUG IT
Octopus V1.1	`stm32f446xx`, new serial after flashing	Pi, direct, **USB 2.0 (black) port**

DEVICE	USB ID	WHERE TO PLUG IT
Cartographer V4	`usb-Cartographer_stm32g431xx_3500350005503155513336 20-if00`	Pi, direct
Tool0 EBB	`usb-Klipper_stm32g0b1xx_420019000F504D4D35313220-if00`	Pi direct, or hub
Tool1 EBB	`usb-Klipper_stm32g0b1xx_590031000D504D4D35313220-if00`	Pi direct, or hub
Nozzle alignment camera	Alcor Micro `058f:5608`	Pi, direct — it wants the bandwidth
BTT-HDMI5 touchscreen	Holtek `04d9:8030`	powered hub

Put the four MCUs and the camera on the Pi's own ports and hang the touchscreen off the powered hub. USB 2.0 ports are preferable for the MCUs — USB 3.0 buys nothing here and its 2.4 GHz noise is a known source of trouble.

> **Do not power the Pi 5 from the Octopus.** The board's Raspberry Pi 5 V header cannot supply what a Pi 5 draws. Use the official 27 W USB-C supply.

The `[mcu]` serial path changes from `/dev/ttyAMA0` to the Octopus's `/dev/serial/by-id/usb-Klipper_stm32f446xx_*-if00`. `scripts/octopus_cutover.sh` reads the real path off the Pi and writes it into `02__Boards_Serials/Mainboard_serial.cfg` for you.

10. Nothing to do on the toolhead

For completeness, these stay exactly as they are and connect to the EBB toolboards, not the Octopus: both extruder motors and their TMC2209 drivers, both hotend heaters and thermistors, part cooling and hotend fans, the X and X1 endstops, the smart filament sensors, and the Cartographer probe body.

11. Before first power-on

- [ ] PSU unplugged from mains, board unpowered.
- [ ] Every driver slot: old jumpers removed, SPI pattern fitted on slots 0–6, slot 7 bare.
- [ ] **All DIAG jumpers removed.**
- [ ] MCU USB-power jumper removed.
- [ ] Each fan header's voltage selector set for the device on it.
- [ ] Polarity checked on `PWR IN`, `MOTOR PWR`, `BED IN`.
- [ ] Drivers fully seated, correct orientation, heatsinks fitted.

First power-on sequence

1. Flash Klipper via SD card: `.9_MCU_Flash/scripts/flash_M1_BTTOctopusV11.sh --build`.
2. Confirm the board enumerates: `ls /dev/serial/by-id/` should show `stm32f446xx`.
3. Bring up on `_deploy/octopus_bringup_minimal.cfg` first — mainboard only, no toolboards.
4. `scripts/octopus_bringup_check.sh` for the automated checks.
5. `DUMP_TMC STEPPER=stepper_x` (then `dual_carriage`, `stepper_y`, `stepper_y1`, `stepper_z`, `stepper_z1`, `stepper_z2`). Real register values prove the SPI jumpers are right; all-zero or all-`ffffffff` means that slot's jumpers are wrong.
6. `QUERY_ENDSTOPS` — trip each switch by hand and confirm the right one changes state.
7. `FORCE_MOVE` each axis a few millimetres to check direction **before** homing anything.

8. Swap to the full `Xplorer_V1.1_IDEX_custom.cfg` and reconnect the toolboards.

Things worth re-checking once it runs

- `rref: 12000` matches BTT TMC2240 modules. A different vendor may use a different reference resistor, and a wrong value silently scales every motor current.
- Run currents were carried over unchanged from the TMC2209 setup. Watch driver temperatures on the first long print — TMC2240 reports its own temperature in `DUMP_TMC`.

Appendix — complete mainboard pin map

PIN	SILKSCREEN	FUNCTION
`PF13` `PF12` `PF14` `PC4`	DRIVER0	`stepper_x` step / dir / enable / CS
`PG0` `PG1` `PF15` `PD11`	DRIVER1	`dual_carriage` step / dir / enable / CS
`PF11` `PG3` `PG5` `PC6`	DRIVER2	`stepper_y` step / dir / enable / CS
`PG4` `PC1` `PA0` `PC7`	DRIVER3	`stepper_y1` step / dir / enable / CS
`PF9` `PF10` `PG2` `PF2`	DRIVER4	`stepper_z` step / dir / enable / CS
`PC13` `PF0` `PF1` `PE4`	DRIVER5	`stepper_z1` step / dir / enable / CS
`PE2` `PE3` `PD4` `PE1`	DRIVER6	`stepper_z2` step / dir / enable / CS
`PA5` `PA6` `PA7`	SPI1	shared TMC2240 SCK / MISO / MOSI
`PG6`	X-STOP	unused
`PG9`	Y-STOP	Y endstop
`PG10`	Z-STOP	Z endstop
`PG11`	Z2-STOP	Z1 endstop
`PG12`	E0DET	Z2 endstop
`PG13`	E1DET	tool offset contact probe
`PG14`	E2DET	T0 buffer runout
`PG15`	E3DET	T1 buffer runout
`PA1`	BED OUT	bed heater
`PF3`	TB	bed thermistor
`PF5`	T1	chamber thermistor
`PA8`	FAN0	case light
`PD12`	FAN2	chamber fan
`PD13`	FAN3	electronics fan
`PD14`	FAN4	T0 buffer supply
`PD15`	FAN5	T1 buffer supply
`PE11`	PS-ON	PSU control